

Jia Yi Ooi

Senior Full-Stack Engineer

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SUMMARY

Senior full-stack engineer, 8+ years, the last 5+ fully remote for teams in Singapore, the US, and Australia. Builds typed UI component systems and the real-time services behind them - most recently the Uchat Business platform, 100k+ messages a day on React/TypeScript and FastAPI on Kubernetes. Available worldwide as a contractor via my own registered entity, or direct with relocation.

LANGUAGES & TECHNOLOGIES

Languages: TypeScript, Python, Elixir, SQL
Frontend: React, Next.js, Vite, TanStack Router, React Query, Socket.IO, Electron, Playwright, Vitest
Backend: Node.js, FastAPI, PostgreSQL, REST API design, load testing
Infrastructure: Kubernetes, Docker, CI/CD (GitHub Actions), Linux, Git
Other: Design systems, real-time systems, Web3 wallets (MetaMask, Tezos)

EXPERIENCE

Career development sabbatical

Apr 2026 – Present

- Built [elixir-opentui](#), [agent-simple-english](#), and [pi-ask-user-question](#) - see Projects.
- Prepare for the Certified Kubernetes Administrator (CKA) exam and deepen Kubernetes and platform tooling.

Uchat

Nov 2024 – Apr 2026

Senior Software Engineer, Business unit

Remote (Singapore)

Business platform of Uchat, a consumer messaging app in Thailand. 500+ companies run their B2C conversations on it.

- Shipped the real-time platform processing 100k+ messages a day for 500+ business accounts - React/TypeScript (Vite, TanStack Router, Socket.IO) over FastAPI on Kubernetes.
- Built the strongly-typed UI component system the platform is assembled from, giving the team a shared vocabulary for new screens.
- Load-tested the chat service to 100k concurrent connections ahead of traffic growth.
- Reviewed the team's frontend work and shaped feature scope directly with product and design.

Accomy

Feb 2024 – Oct 2024

Senior Software Engineer

Remote (Singapore)

- Led 2 frontend engineers to ship the MVP of a corporate hotel booking platform in 4 months, from empty repo to launch.
- Established the Next.js + TypeScript codebase from scratch, mentored 2 engineers onto it, and drove architecture decisions with product and design.

Coinalpha (Hummingbot)

Jun 2021 – Nov 2023

Software Engineer

Remote (US)

Hummingbot is an open-source market-making framework for crypto trading bots.

- Built the React + FastAPI liquidity mining platform used by ~1,000 bot operators facilitating ~\$100k in daily trading volume.
- Built and published [@hummingbot/hbui](#), the Hummingbot design system - design tokens synced from Figma into a Tailwind + React package consumed across the organisation's frontends.
- Fixed memory leaks that exhausted 8 GB machines within hours, cutting steady-state memory ~90%.

Areca System

2020 – 2021

Contract Software Engineer

Penang, Malaysia

- Built an Electron point-of-sale system from scratch for a 30-store supermarket chain.

Aglive

2017 – 2020

Software Engineer

Remote (Australia)

- Built and operated the tracking dashboard covering 500+ trucks and buses on the Austracker platform.
- Led deduplication of multi-million-row GPS data, removing ~25% duplicate records that fed customer reports.

PROJECTS

- [elixir-opentui](#) - terminal UI framework for Elixir with Zig NIF backend: Elm-inspired architecture, declarative View DSL, 15+ widgets, double-buffered rendering.
- [agent-simple-english](#) - deterministic ASD-STE100 linter: Engine, CLI, and pi/Claude Code adapters that gate writes, edits, commits, and replies.
- [pi-ask-user-question](#) - pi TUI extension for structured questions: multi-select, free-text Other, TUI-only with transcript rendering.

EDUCATION

B.IT Communications & Networking (Hons), Universiti Tunku Abdul Rahman

May 2017